

3-Tier Architecture with Application Load Balancer

Project: 3-Tier Architecture with Application Load Balancer (AWS)

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Project Overview:

This project demonstrates the deployment of a **3-Tier Architecture** on AWS using an **Application Load Balancer (ALB)**. The architecture separates the application into three independent layers to improve scalability, security, and maintainability.

- Architecture Overview:**
- Web Server Layer: Nginx (EC2 instance with Public subnet)
 - Application Server Layer: Tomcat (EC2 instance with Public subnet)
 - Database Server Layer: MySQL (EC2 instance with Public subnet)
 - Application Load Balancer

Steps to create a 3-Tier Architecture:

Step1: Create a Virtual Private Cloud (VPC)

A Virtual Private Cloud (VPC) is created with a custom CIDR block. AWS automatically creates the default Route Table, Network ACL, and Security Group for the VPC.

Your VPCs (1) Info

Last updated about 3 hours ago

Actions

Create VPC

Find VPCs by attribute or tag

< 1 >

☐	Name	VPC ID	State	Encryption c...	Encryption contr...
☐	3-tierwith-lb	vpc-0b45e09aaeefe8ba0	Available	-	-

Result:

Custom VPC Created Successfully.

Step2: Create Subnets

Three public subnets are created across different Availability Zones to improve availability and fault tolerance.

- Subnet 1 – Web Server
- Subnet 2 – Application Server
- Subnet 3 – Database Server

Subnets (3) Info

Last updated about 3 hours ago

Actions

Create subnet

Find subnets by attribute or tag

< 1 >

☐	Name	Subnet ID	State	VPC
☐	subnet-1	subnet-02f89b4af7b1b2334	Available	vpc-0b45e09aaeefe8ba0 3-tie...
☐	subnet-2	subnet-04e0f1db02868235c	Available	vpc-0b45e09aaeefe8ba0 3-tie...
☐	subnet-3	subnet-0b6e11a817e09667a	Available	vpc-0b45e09aaeefe8ba0 3-tie...

Result:

Three subnets were successfully deployed across multiple Availability Zones.

Step3: Create and Attach an Internet Gateway

An Internet Gateway is created and attached to the VPC, enabling internet access for resources deployed in public subnets.

Internet gateways (1) Info

Actions

Create internet gateway

Find internet gateways by attribute or tag

< 1 >

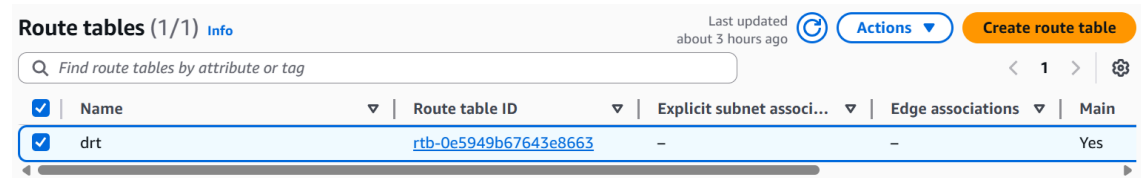
☐	Name	Internet gateway ID	State	VPC ID
☐	ig	igw-0f19e3f33bcf46571	Attached	vpc-0b45e09aaeefe8ba0 3-tie

Result:

Internet Gateway attached successfully.

Step4: Configure the Route Table

The default Route Table is updated with a route (0.0.0.0/0) pointing to the Internet Gateway, allowing outbound internet connectivity.

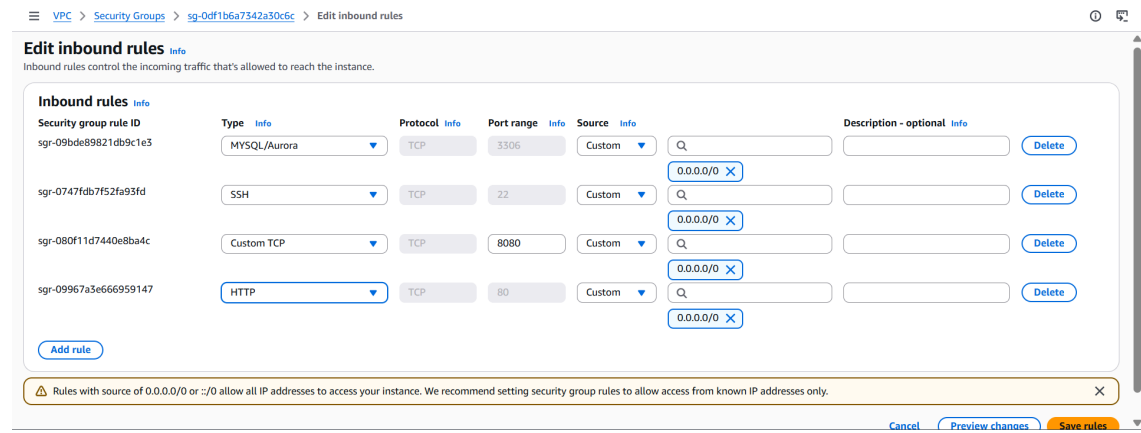


Result:

Public internet access is configured successfully.

Step5: Configure the Security Group

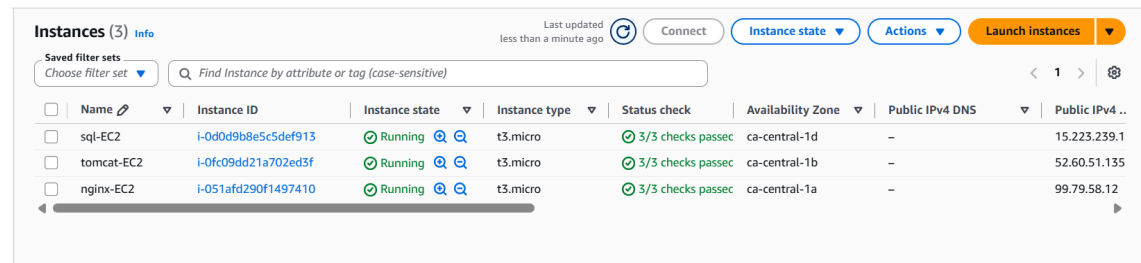
A Security Group is configured with the required inbound rules.



Step6: Launch an EC2 Instance

Three EC2 instances are provisioned using Terraform.

Instances	Purpose
Nginx EC2	Web Server
Tomcat EC2	Application Server
MySQL EC2	Database Server



Result:

All three EC2 instances launched successfully.

Step7: Install and Configure Nginx

Connect to the web server with MobaXterm

root@ip-10-1-1-219:/home/ubuntu#

cmd: apt update

cmd: apt install nginx -y

Now we check if Nginx is installed on our web server.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Nginx is installed on our web server.

Step8: Install Apache Tomcat

Connect to app-ser with MobaXterm

```
cmd: sudo apt update
cmd: apt install default-jdk -y
cmd: wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.49/bin/apache-tomcat-11.0.14.tar.gz
cmd: ls -lt
cmd: tar -xvzf apache-tomcat-11.0.14.tar.gz
cmd: ls -lt
cmd: mv apache-tomcat-11.0.14 tomcat
cmd: ls
cmd: cd tomcat/bin/ or cd /home/ubuntu/tomcat/bin
cmd: ls
cmd: ./startup.sh
```

Now we check if Tomcat is installed or not on our app server.

← → ↻ ⚠ Not secure 3.99.216.66:8080 ☆ 📁 📄 🌐 ⋮
📁 All Bookmarks

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Apache Tomcat/10.1.56

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

- [Tomcat Setup](#)
- [First Web Application](#)
- [Realms & AAA](#)
- [JDBC DataSources](#)
- [Examples](#)
- [Servlet Specifications](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 10.1 access to the manager

Documentation

- [Tomcat 10.1 Documentation](#)
- [Tomcat 10.1 Configuration](#)
- [Tomcat Wiki](#)

Getting Help

[FAQ](#) and [Mailing Lists](#)

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security

Tomcat is installed in our app server.

Step9: Install MySQL

Connect to db-ser with MobaXterm

```
Cmd: apt update
Cmd: apt install mysql-server -y
Cmd: mysql_secure_installation
➤ Edit the MySQL configuration file using a text editor:
# vi /etc/mysql/mysql.conf.d/mysqld.cnf
➤ Locate the following line: bind-address = 127.0.0.1
• bind-address = 0.0.0.0
• Save and exit (CTRL + X, then Y, then Enter).
Cmd: systemctl restart mysql
```

Cmd: systemctl status mysql

```
Escape character is '^['.
Connection closed by foreign host.
root@ip-10-1-3-39:/home/ubuntu# vi etc/mysql/mysql.conf.d/mysql.cnf
root@ip-10-1-3-39:/home/ubuntu# vi etc/mysql/mysql.conf.d/mysqld.cnf
root@ip-10-1-3-39:/home/ubuntu# vi /etc/mysql/mysql.conf.d/mysqld.cnf
root@ip-10-1-3-39:/home/ubuntu# systemctl restart mysql
root@ip-10-1-3-39:/home/ubuntu# systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-07-06 06:05:26 UTC; 10s ago
 Invocation: 69cf157935b043c1928faa6f2529b94a
   Process: 15734 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0)
  Main PID: 15751 (mysqld)
    Status: "Server is operational"
     Tasks: 35 (limit: 627)
    Memory: 474.8M (peak: 474.8M)
       CPU: 845ms
    CGroup: /system.slice/mysql.service
            └─15751 /usr/sbin/mysqld

Jul 06 06:05:25 ip-10-1-3-39 systemd[1]: Starting mysql.service - MySQL Community Server...
Jul 06 06:05:26 ip-10-1-3-39 systemd[1]: Started mysql.service - MySQL Community Server.
```

MySQL is installed on our dbserver.

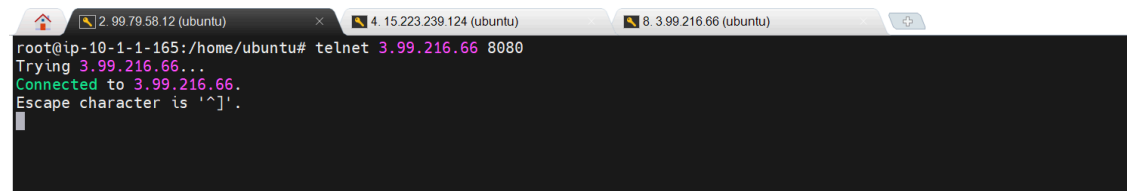
Step10: Verify Server Connectivity

Now connect the web server to the app server.

Open the web server in MobaXterm

Cmd: telnet 3.99.216.66 8080

In this, 3.99.216.66 is the app server's public IP. 8080 is the HTTP port number



```
root@ip-10-1-1-165:/home/ubuntu# telnet 3.99.216.66 8080
Trying 3.99.216.66...
Connected to 3.99.216.66.
Escape character is '^['.
```

Our web server is successfully connected to the app server.

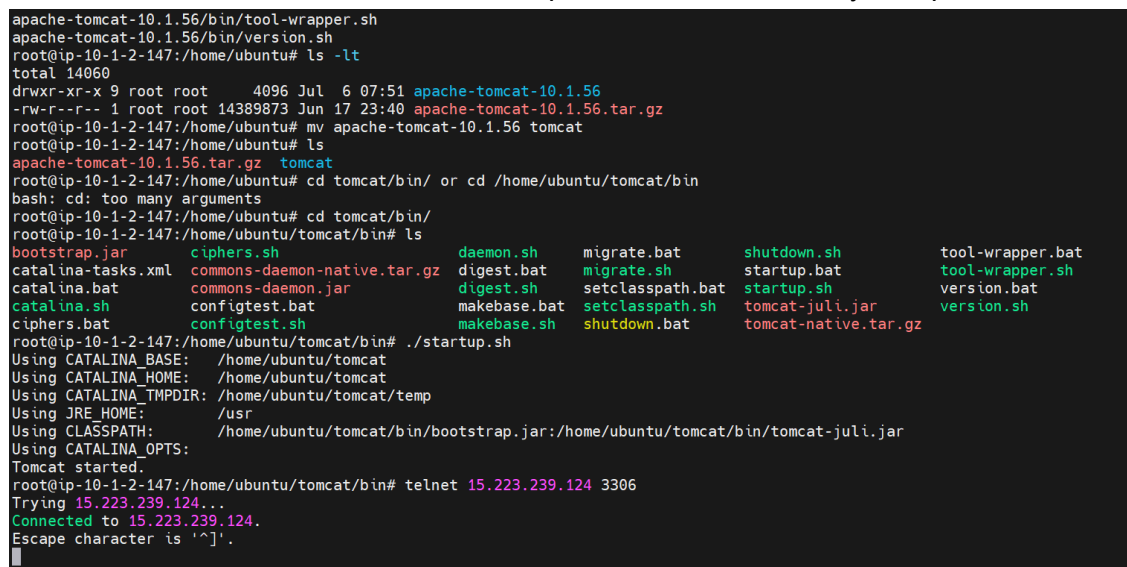
Step11: Verify Server Connectivity

Now connect the app server to the db server.

Open appserver in MobaXterm

Cmd: telnet 15.223.239.124 3306

In this, 15.223.239.124 is the DB server's public IP. 3306 is the MySQL port number.



```
apache-tomcat-10.1.56/bin/tool-wrapper.sh
apache-tomcat-10.1.56/bin/version.sh
root@ip-10-1-2-147:/home/ubuntu# ls -lt
total 14060
drwxr-xr-x 9 root root 4096 Jul 6 07:51 apache-tomcat-10.1.56
-rw-r--r-- 1 root root 14389873 Jun 17 23:40 apache-tomcat-10.1.56.tar.gz
root@ip-10-1-2-147:/home/ubuntu# mv apache-tomcat-10.1.56 tomcat
root@ip-10-1-2-147:/home/ubuntu# ls
apache-tomcat-10.1.56.tar.gz tomcat
root@ip-10-1-2-147:/home/ubuntu# cd tomcat/bin/ or cd /home/ubuntu/tomcat/bin
bash: cd: too many arguments
root@ip-10-1-2-147:/home/ubuntu# cd tomcat/bin/
root@ip-10-1-2-147:/home/ubuntu/tomcat/bin# ls
bootstrap.jar  ciphers.sh  commons-daemon-native.tar.gz  daemon.sh  migrate.bat  shutdown.sh  tool-wrapper.bat
catalina-tasks.xml  commons-daemon.jar  digest.sh  migrate.sh  setclasspath.bat  startup.bat  tool-wrapper.sh
catalina.bat  configtest.bat  makebase.bat  setclasspath.sh  tomcat-juli.jar  version.bat  version.sh
ciphers.bat  configtest.sh  makebase.sh  shutdown.bat  tomcat-native.tar.gz
root@ip-10-1-2-147:/home/ubuntu/tomcat/bin# ./startup.sh
Using CATALINA_BASE: /home/ubuntu/tomcat
Using CATALINA_HOME: /home/ubuntu/tomcat
Using CATALINA_TMPDIR: /home/ubuntu/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /home/ubuntu/tomcat/bin/bootstrap.jar:/home/ubuntu/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@ip-10-1-2-147:/home/ubuntu/tomcat/bin# telnet 15.223.239.124 3306
Trying 15.223.239.124...
Connected to 15.223.239.124.
Escape character is '^['.
```

Step12: Verify Server Connectivity

Now check whether the DB server is connected to the app server or not.

Open dbserver in MobaXterm

Cmd: telnet 3.99.216.66 8080

Step13:Create Target Group

Nginx Target Group (Port 80)

Filter target groups							
	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
	nginx-tg	arn:aws:elasticloadbalanc...	80	HTTP	Instance	alb	vpc-0b45e09aaefe8

Step14: Create Target Group

Tomcat Target Group (Port 8080)

	tomcat-tg	arn:aws:elasticloadbalanc...	80	HTTP	Instance	alb-1	vpc-0b45e09aaefe8
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Step15: Create Application Load Balancer

Application Load Balancers are deployed across all three public subnets. Each ALB is associated with the default Security Group and configured to forward traffic to its corresponding Target Group.

	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
	alb	Active	application	Internet-facing	IPv4	vpc-0b45e09aaefe8ba0	3 Availability Zones

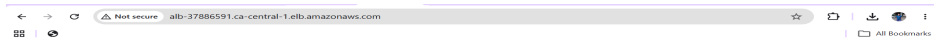
Step16: Create Other Application Load Balancer

Application Load Balancers are deployed across all three public subnets. Each ALB is associated with the default Security Group and configured to forward traffic to its corresponding Target Group.

	alb-1	Active	application	Internet-facing	IPv4	vpc-0b45e09aaefe8ba0	3 Availability Zones
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The Final result is

Copy the First Application Load Balancer DNS and paste it into Chrome.



Copy the Second Application Load Balancer DNS and paste it into Chrome.

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Apache Tomcat/11.0.23ASF

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:
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[Manager Application](#) [How-To](#)
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Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:
`$CATALINA_HOME/conf/tomcat-users.xml`
In Tomcat 11.0 access to the manager application is split between different users.
[Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 11.0 Documentation](#)
[Tomcat 11.0 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:
`$CATALINA_HOME/RUNNING.txt`
Developers may be interested in:
[Tomcat 11.0 Bug Database](#)
[Tomcat 11.0 JavaDocs](#)
[Tomcat 11.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:
[tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
[tomcat-users](#)
User support and discussion
[tomcat-dev](#)
User support and discussion for [Apache Tomcat](#)
[tomcat-dev](#)
Development mailing list, including commit messages

Other Downloads

[Tomcat Connectors](#)
[Tomcat Native](#)
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Conclusion:

This project successfully demonstrates the deployment of a scalable 3-tier architecture on AWS using Terraform. The infrastructure includes a custom VPC, public subnets, EC2 instances for the Web, Application, and Database tiers, Application Load Balancers, Target Groups, and Listeners.